ANSWERS

1.

Question	Answer	Marks
6(a)	(hard) steel	**B1**
6(b)	(place compass near to a pole and) mark a dot at further end of needle	**B1**
	move compass so the first end coincides with the dot **and** mark second dot	**B1**
	repeat (many times) **and** join up dots	**B1**
6(c)	magnetism <u>induced</u> in iron bar **or** poles <u>induced</u> on iron bar	**B1**
	end P becomes an N-pole	**B1**
	unlike poles attract **or** (force of) attraction greater than friction / repulsion	**B1**

2.

(a) **(i)** **horizontal** arrow to right (by eye) [B1]

(ii) forces / resultant causes moment or (turns because) force is not at pivot [B1]

(b) mark made at one end/pole/direction of compass (on paper) [B1]
move compass so that other end of compass is on mark **and** remark [B1]
join marks made as compass moved on in some way (to draw line) [B1]

3.

(a) steel/alnico/SmCo/NdFeB/magnetite B1

(b) one needle fully correct **or** both angles correct – i.e. A bottom left to top right diagonal ($0 < \text{angle} < 90°$) **and** B horizontal C1
both needles fully correct (fully = angle and orientation) A1

(c) (place) magnet in solenoid B1
a.c. supply to solenoid/coil (ignore cell/battery symbol) B1
withdraw magnet (slowly) **or** reduce current (slowly) B1